

The Spectrum of CAR Cellular Effectors: Modes of Action in Anti-Tumor Immunity.

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Chimeric antigen receptor-T cells have spearheaded the field of adoptive cell therapy and have shown remarkable results in treating hematological neoplasia. Because of the different biology of solid tumors compared to hematological tumors, response rates of CAR-T cells could not be transferred to solid entities yet. CAR engineering has added co-stimulatory domains, transgenic cytokines and switch receptors to improve performance and persistence in a hostile tumor microenvironment, but because of the inherent cell type limitations of CAR-T cells, including HLA incompatibility, toxicities (cytokine release syndrome, neurotoxicity) and high costs due to the logistically challenging preparation process for autologous cells, the use of alternative immune cells is gaining traction. NK cells and $\gamma\delta$ T cells that do not need HLA compatibility or macrophages and dendritic cells with additional properties such as phagocytosis or antigen presentation are increasingly seen as cellular vehicles with potential for application. As these cells possess distinct properties, clinicians and researchers need a thorough understanding of their peculiarities and commonalities. This review will compare these different cell types and their specific modes of action seen upon CAR activation.

Challenges and Recent Advances in NK Cell-Targeted Immunotherapies in Solid Tumors.

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Natural killer (NK) cell is a powerful malignant cells killer, providing rapid immune responses via direct cytotoxicity without the need of antigen processing and presentation. It plays an essential role in preventing early tumor, metastasis and minimal residual disease. Although adoptive NK therapies achieved great success in clinical trials against hematologic malignancies, their accumulation, activation, cytotoxic and immunoregulatory functions are severely impaired in the immunosuppressive microenvironment of solid tumors. Now with better understandings of the tumor evasive mechanisms from NK-mediated immunosurveillance, immunotherapies targeting the key molecules for NK cell dysfunction and exhaustion have been developed and tested in both preclinical and clinical studies. In this review, we introduce the challenges that NK cells encountered in solid tumor microenvironment (TME) and the therapeutic approaches to overcome these limitations, followed by an outline of the recent preclinical advances and the latest clinical outcomes of NK-based immunotherapies, as well as promising strategies to optimize current NK-targeted immunotherapies for solid tumors.

What Inhibits Natural Killers' Performance in Tumour.

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Natural killer cells are innate lymphocytes with the ability to lyse tumour cells depending on the balance of their activating and inhibiting receptors. Growing numbers of clinical trials show promising results of NK cell-based immunotherapies. Unlike T cells, NK cells can lyse tumour cells independent of antigen presentation, based simply on their activation and inhibition receptors. Various strategies to improve NK cell-based therapies are being developed, all with one goal: to shift the balance to activation. In this review, we discuss the current understanding of ways NK cells can lyse tumour cells and all the inhibitory signals stopping their cytotoxic potential.

Next Generation Immuno-Oncology Strategies: Unleashing NK Cells Activity.

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In recent years, immunotherapy has become a powerful therapeutic option against multiple malignancies. The unique capacity of natural killer (NK) cells to attack cancer cells without antigen specificity makes them an optimal immunotherapeutic tool for targeting tumors. Several approaches are currently being pursued to maximize the anti-tumor properties of NK cells in the clinic, including the development of NK cell expansion protocols for adoptive transfer, the establishment of a favorable microenvironment for NK cell activity, the redirection of NK cell activity against tumor cells, and the blockage of inhibitory mechanisms that constrain NK cell function. We here summarize the recent strategies in NK cell-based immunotherapies and discuss the requirement to further optimize these approaches for enhancement of the clinical outcome of NK cell-based immunotherapy targeting tumors.

Potential therapeutic role of natural killer cells in cancer.

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Cancer immunotherapy is a growing field that aims at restoring and enhancing immune function to combat oncogenic conditions. One target of this field is natural killer (NK) cells. Part of innate immunity, NK cells are able to kill tumor cells without previous priming. Results from stem cell transplants containing alloreactive donor NK cells and in vitro work have evidenced a great antitumor potential. In addition, NK cells are likely to interact with dendritic cells, potent antigen-presenting cells, thus forming a bridge between innate and adaptive immunity. This review aims to provide an overview of NK cells with particular emphasis on properties that can and are being targeted in order to potentiate the antitumor activity of these cells.

Epidermal growth factor receptor: a re-emerging target in glioblastoma.

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Amplification and overexpression of the epidermal growth factor receptor (EGFR) gene are a hallmark of primary glioblastoma (45%), making it a prime target for therapy. In addition, these amplifications are frequently associated with oncogenic mutations in the extracellular domain. However, efforts at targeting the EGFR tyrosine kinase using small molecule inhibitors or antibodies have shown disappointing efficacy in clinical trials for newly diagnosed or recurrent glioblastoma. Here, we review recent insights into molecular mechanisms relevant for effective targeting of the EGFR pathway. Molecular workup of glioblastoma tissue of patients under treatment with small molecule inhibitors has established drug concentrations in the tumor tissue, and has shed light on the effectiveness of target inhibition and respective effects on pathway signaling. Further, functional analyses of interaction of small molecule inhibitors with distinct properties to bind to the active or inactive form of EGFR have provided new insights that will impact the choice of drugs. Finally, vaccination approaches targeting the EGFRvIII mutant featuring a tumor-specific antigen have shown promising results that warrant larger controlled clinical trials. A combination of preclinical and clinical studies at the molecular level has provided new insights that will allow refining strategies for targeting the EGFR pathway in glioblastoma.

Primary glioblastomas with and without EGFR amplification: relationship to genetic alterations and clinicopathological features.

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Glioblastomas express a notable heterogeneity in both the histological and cell patterns with glial astrocytic differentiation. Primary glioblastoma, which is the most frequent presentation (90-95%), occurs mainly in older patients and arises de novo, without any clinical or histological evidence of a less malignant precursor lesion. EGFR amplification has been identified as a genetic hallmark of primary glioblastomas and occurs in 40-60% of cases. However, there exist primary glioblastomas without EGFR amplification/overexpression. The purpose of this study was to stabilize the association between cases with and without EGFR gene amplification with clinical and genetic parameters in 45 cases of primary glioblastomas. EGFR amplification was observed in 24 cases (53%), while in the remaining 21 cases (47%) this alteration was not displayed. And whereas EGFR was overexpressed in 79% of cases with EGFR amplification, only 33% of the cases without EGFR amplification showed overexpression. The amplification of EGFR was associated with amplifications in MDM2 and CDK4 and a higher percentage of cases with promoter methylation of INK4a. Only one case of glioblastoma with EGFR amplification presented TP53 mutation simultaneously. Seven remaining cases with TP53 mutations were glioblastomas without EGFR amplification. The INK4a, INK4b and ARF deletions were similar in the two groups. Primary glioblastomas with and without EGFR amplification did not show any significant differences in average survival. The genetic studies suggest the existence of molecular subtypes within primary glioblastoma that may, when fully defined, contribute toward the development of drugs that specifically target tumors with divergent genetic profiles.

Treatment of primary glioblastoma multiforme with cetuximab, radiotherapy and temozolomide (GERT)--phase I/II trial: study protocol.

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The implementation of combined radiochemotherapy (RCHT) with temozolomide (TMZ) has lead to a significant increase in overall survival times in patients with Glioblastoma multiforme (GBM), however, outcome still remains unsatisfactory. The majority of GBMs show an overexpression and/or amplification of the epidermal growth factor receptor (EGFR). Therefore, addition of EGFR-inhibition with cetuximab to the current standard treatment approach with radiotherapy and TMZ seems promising. GERT is a one-armed single-center phase I/II trial. In a first step, dose-escalation of TMZ from 50 mg/m² to 75 mg/m² together with radiotherapy and cetuximab will be performed. Should safety be proven, the phase II trial will be initiated with the standard dose of 75 mg/m² of TMZ. Cetuximab will be applied in the standard application dose of 400 mg/m² in week 1, thereafter at a dose of 250 mg/m² weekly. A total of 46 patients will be included into this phase I/II trial. Primary endpoints are feasibility and toxicity, secondary endpoints are overall and progression-free survival. An interim analysis will be performed after inclusion of 15 patients into the main study. Patients' enrollment will be performed over a period of 2 years. The observation time will end 2 years after inclusion of the last patient. The goal of this study is to evaluate the safety and efficacy of combined RCHT-immunotherapy with TMZ and cetuximab as first-line treatment for patients with primary GBM.

Molecular pathogenesis of malignant gliomas.

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De novo glioblastomas develop in older patients without prior clinical history of less malignant tumors. Progressive glioblastomas are common among younger patients and arise through progression from lower-grade astrocytomas. CDKN2A deletions, PTEN alterations, and EGFR amplification are more prevalent among de novo glioblastomas, whereas p53 mutations are more common among progressive glioblastomas. Loss of heterozygosity (LOH) for chromosome 10 is seen uniformly among both de novo and progressive high-grade astrocytomas. The inactivation of the PTEN gene is found in approximately 30% to 40% of astrocytomas with chromosome 10 loss, and LOH pattern in the remaining astrocytomas strongly supports the presence of another yet unidentified tumor suppressor gene telomeric to PTEN. More than 80% of oligodendrogliomas exhibit LOH for 1p and 19q alleles. Oligoastrocytomas with 1p/19q LOH are related to oligodendrogliomas, and those with p53 mutations are related to astrocytomas.

Genetics of Glioblastoma in Moroccan population: Review of literature.

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Glioblastoma is the most aggressive malignant type of central nervous system tumors. The literature review revealed that the most common genetic abnormalities in primary and secondary glioblastomas are IDH1 / 2 mutations, p53 mutations, and overexpression of EGFR. To our knowledge, this is the first Moroccan study to provide a global picture of genetic studies performed on Moroccan patients to describe genetic markers and their frequency in our population. An in-depth research in ScienceDirect and Pubmed on glioblastoma articles treated in Morocco, this research is based on the following keywords: glioblastoma in Morocco, primary glioblastoma in Morocco, secondary glioblastoma in Morocco. We found only three research articles on genomic alteration of IDH1 / 2, p53 and EGFR expression. The frequency of these gene mutations in our population was similar to those reported from other populations. Additional molecular studies need to be performed on other genes to describe other genetic markers in the Moroccan population.